cBioPortal
FOR CANCER GENOMICS

Tutorial 2
Single Study Query

Query one or more genes in a single study

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cbioportal.org/tutorials

Tutorial objectives

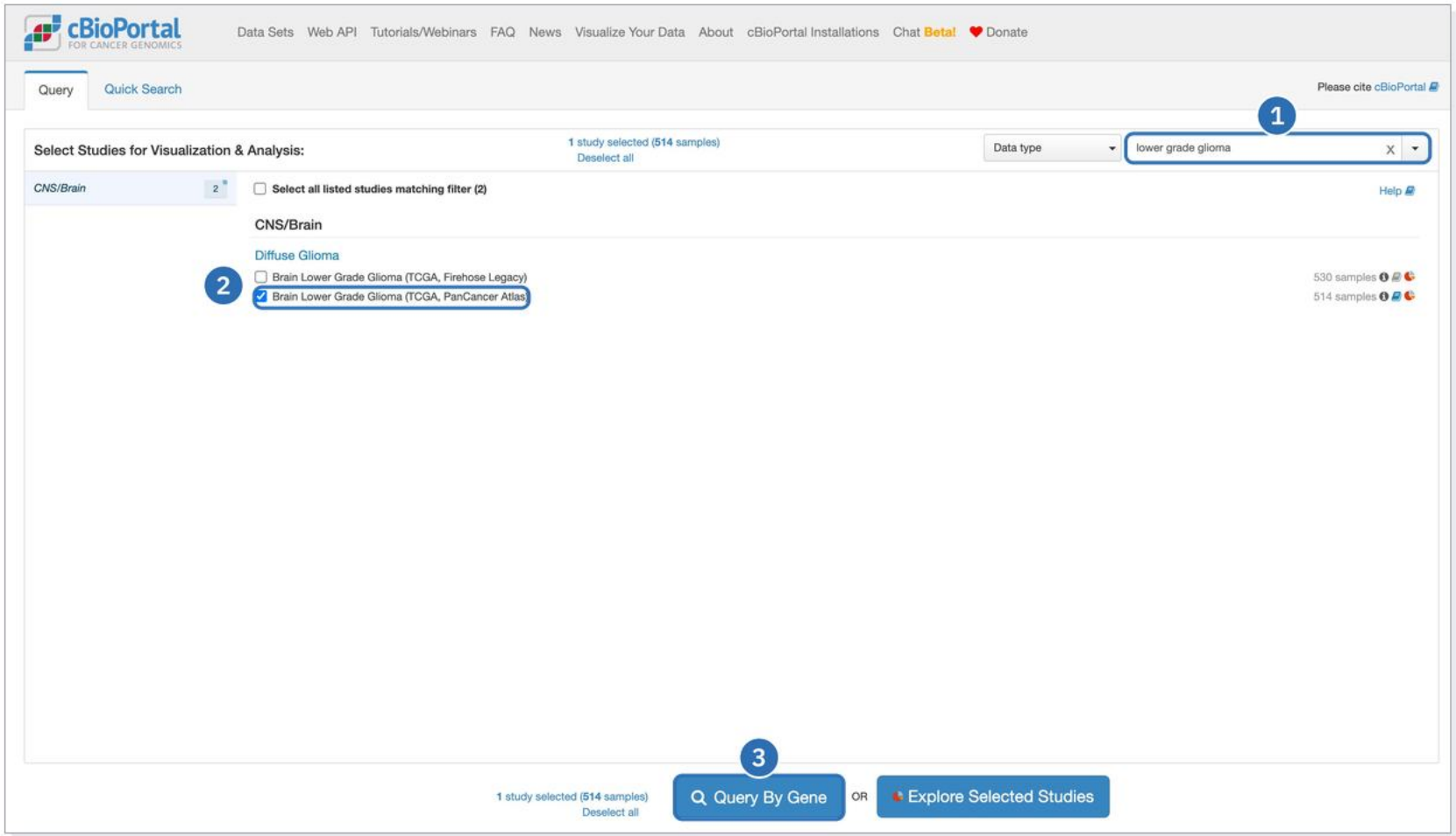
- Run a query from the main page: choose a study, genomic profiles, samples and genes
- Read the Results View header, change its settings, and modify a query
- Walk through the Results View tabs: OncoPrint, Cancer Types Summary, Mutual Exclusivity, Plots, Mutations, Structural Variants, Co-expression, Comparison/Survival, CN Segments, Pathways and Download
- Answer biological questions along the way (Q and A, with green numbers)

Overview of the Results View tabs

Which tabs you see depends on the query and on the data in the study: e.g. a study without mRNA expression data has no Co-expression tab.

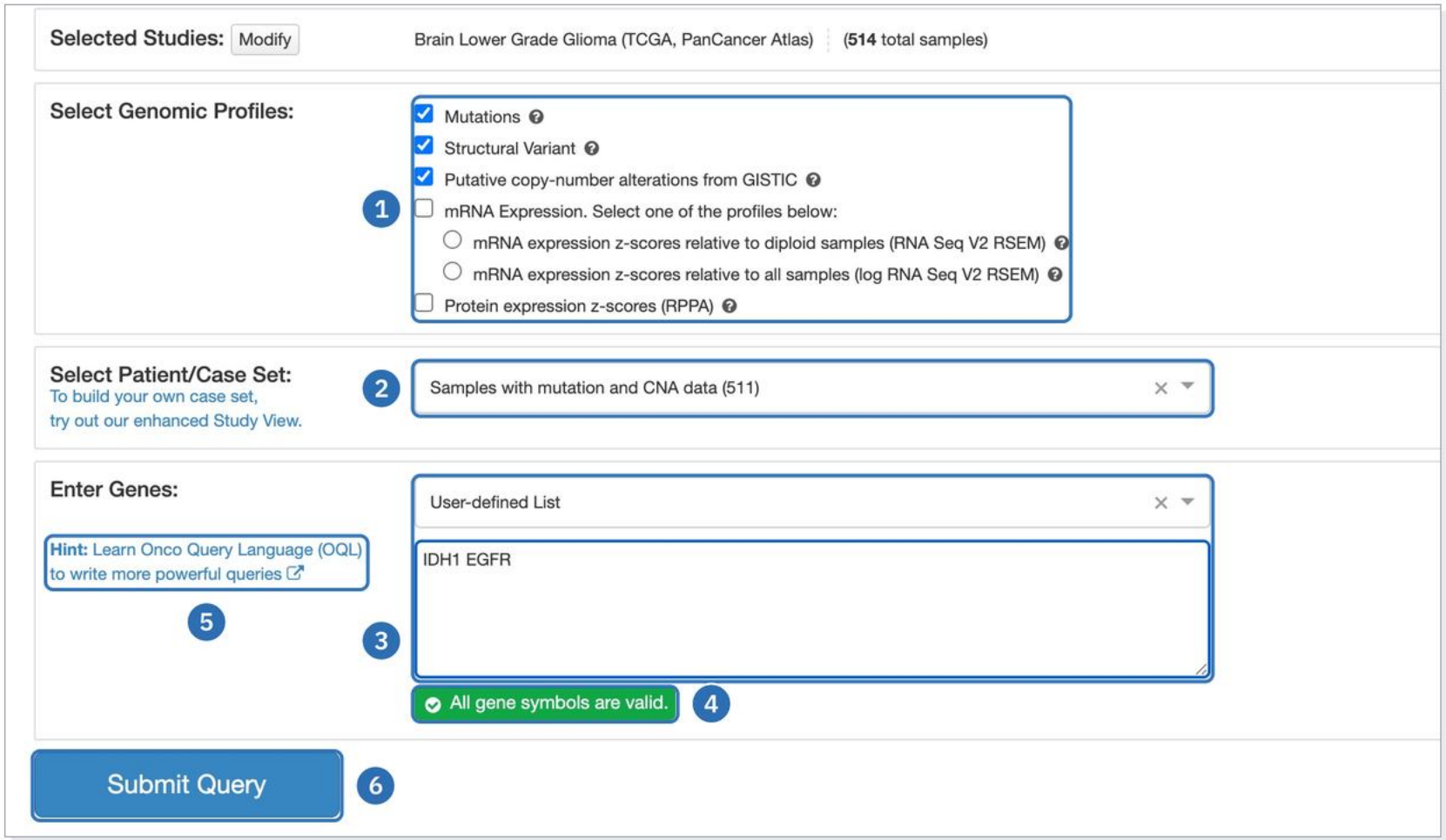
- OncoPrint: an overview of the genetic alterations in each query gene, across the queried samples
- Cancer Types Summary: how often each query gene is altered in each cancer type of the study
- Mutual Exclusivity: tests whether pairs of query genes tend to be altered in different samples, or in the same ones
- Plots: the relationships among genetic alterations, gene expression, protein levels, DNA methylation and clinical attributes
- Mutations: the mutations in each query gene, along the protein and in a table
- Structural Variants: fusions and other structural variants in each query gene
- Co-expression: the genes whose mRNA or protein levels correlate with each query gene
- Comparison/Survival: overlap, outcomes, clinical attributes and genomic data, compared between groups of samples defined by the query
- CN Segments: copy-number changes in the Integrative Genomics Viewer (IGV)
- Pathways: the query genes in known pathways, from PathwayMapper and NDEx
- Download: download the data, or copy sample lists

Starting a query from the main page



1. Find your study of interest: type part of its name in the search box. In this example, “lower grade glioma”.
2. Tick your study of interest. In this example, Brain Lower Grade Glioma (TCGA, PanCancer Atlas), LGG for short.
3. Click “Query By Gene”. (“Explore Selected Studies” opens the Study View instead: see Tutorial 1.)

The query form

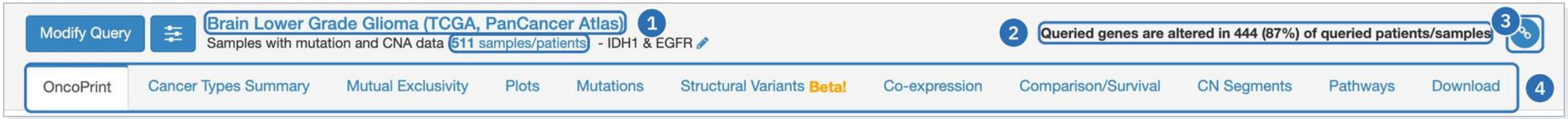


1. Choose the genomic profiles. Mutations, structural variants and copy number are ticked by default; add mRNA or protein expression if you need them.
2. Choose the samples. A set that matches the profiles is picked for you (here, 511 samples with mutation and CNA data).
3. Type gene symbols, or pick a predefined gene list from the menu. Here, IDH1 EGFR. Not case sensitive; separate genes with spaces, commas or new lines.
4. The symbols are checked as you type.
5. Onco Query Language (OQL) narrows a query to specific alterations, e.g. EGFR: AMP MUT. See Tutorial 5.
6. Click “Submit Query”.

The Results View

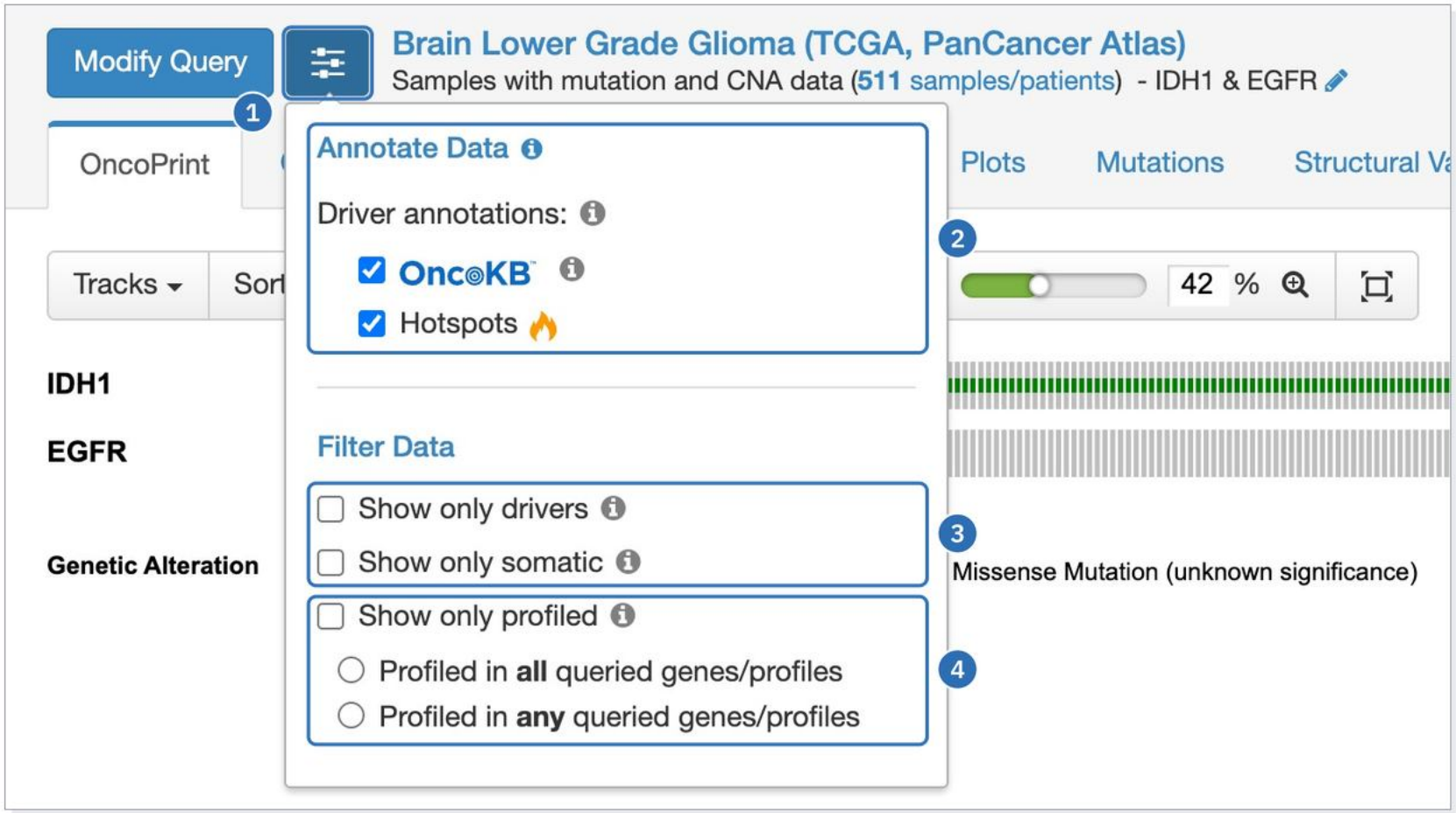
- Submitting a query, here or from the Study View (end of Tutorial 1), opens the Results View.
- Its tabs analyse the query genes in the queried samples. They all share one header.

Results View: the header



1. The study and the queried samples (511). Click the study name to open the whole study in the Study View, or the sample count to open the Study View for just these samples.
2. How many queried samples have an alteration in any query gene: 444 (87%) for IDH1 and EGFR.
3. Copy a link to this query, to share it or come back to it later.
4. The tabs. The next slides go through each of them.

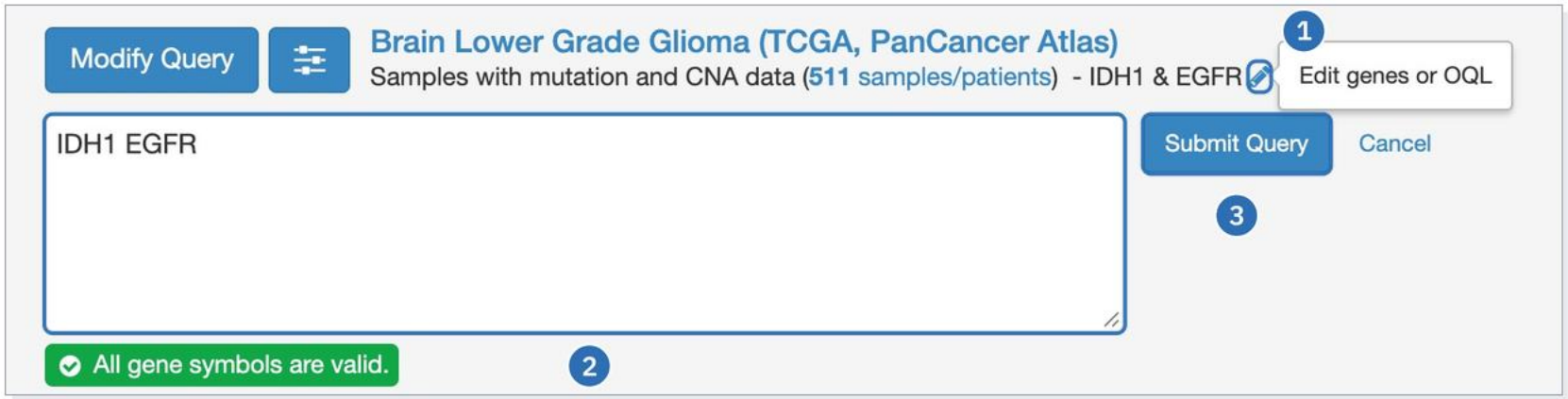
Results View: settings



1. The settings menu controls which alterations are included, on every tab.
2. "Annotate Data": OncoKB and Hotspots annotate which mutations are putative drivers and which are variants of unknown significance (VUS).
3. "Show only drivers" or "Show only somatic": the other alterations are then treated as no alteration.
4. "Show only profiled": leave out samples in which the query genes were not profiled (in all, or in any, of the profiles).

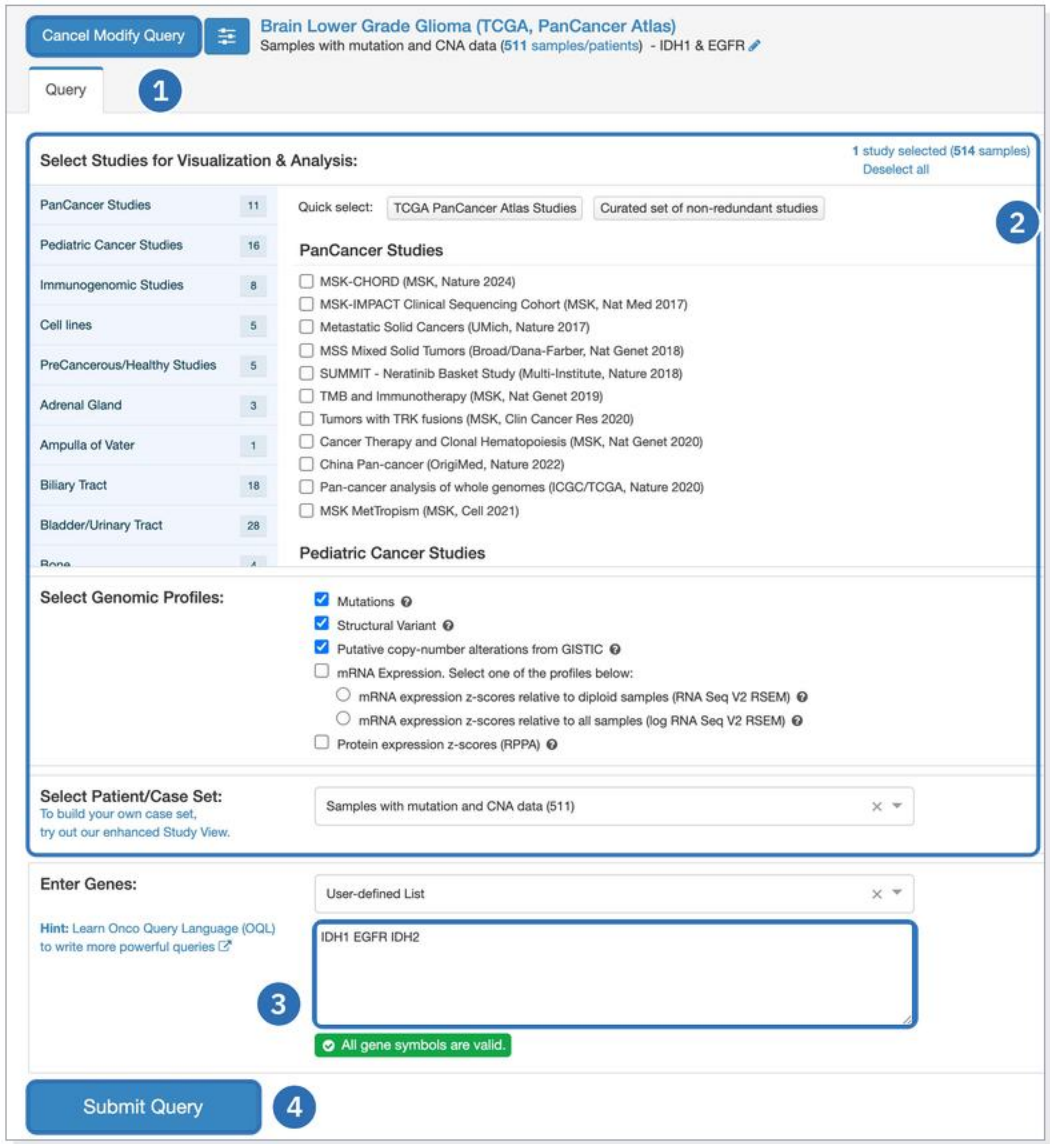
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Editing the genes



1. Click the pencil next to the genes for a quick edit.
2. Change the genes or the OQL here.
3. Click “Submit Query” to rerun the query with the new genes. Everything else (study, profiles, samples) stays the same; to change those, use “Modify Query”.

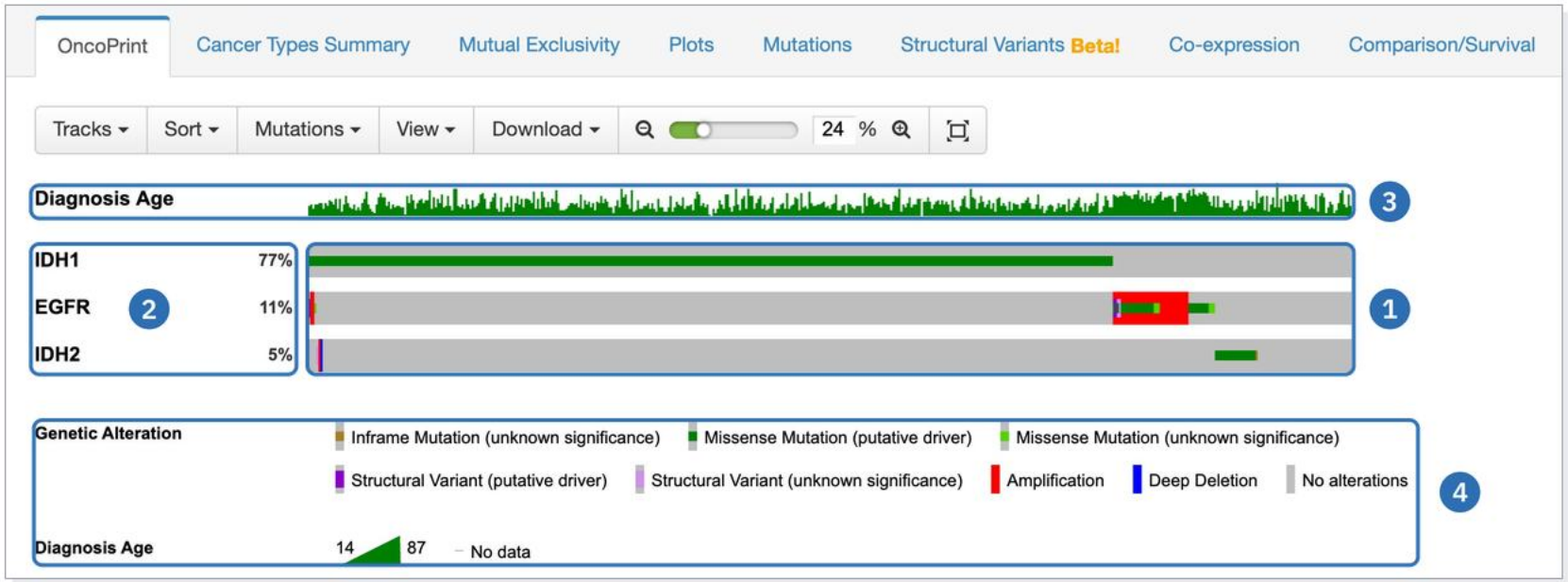
Modify Query



1. "Modify Query", on every tab, opens the query form with the current query filled in.
2. Change the study, the profiles or the samples.
3. Or change the genes: here we add a third gene, IDH1 EGFR IDH2.
4. Click "Submit Query". The rest of this tutorial uses this query.

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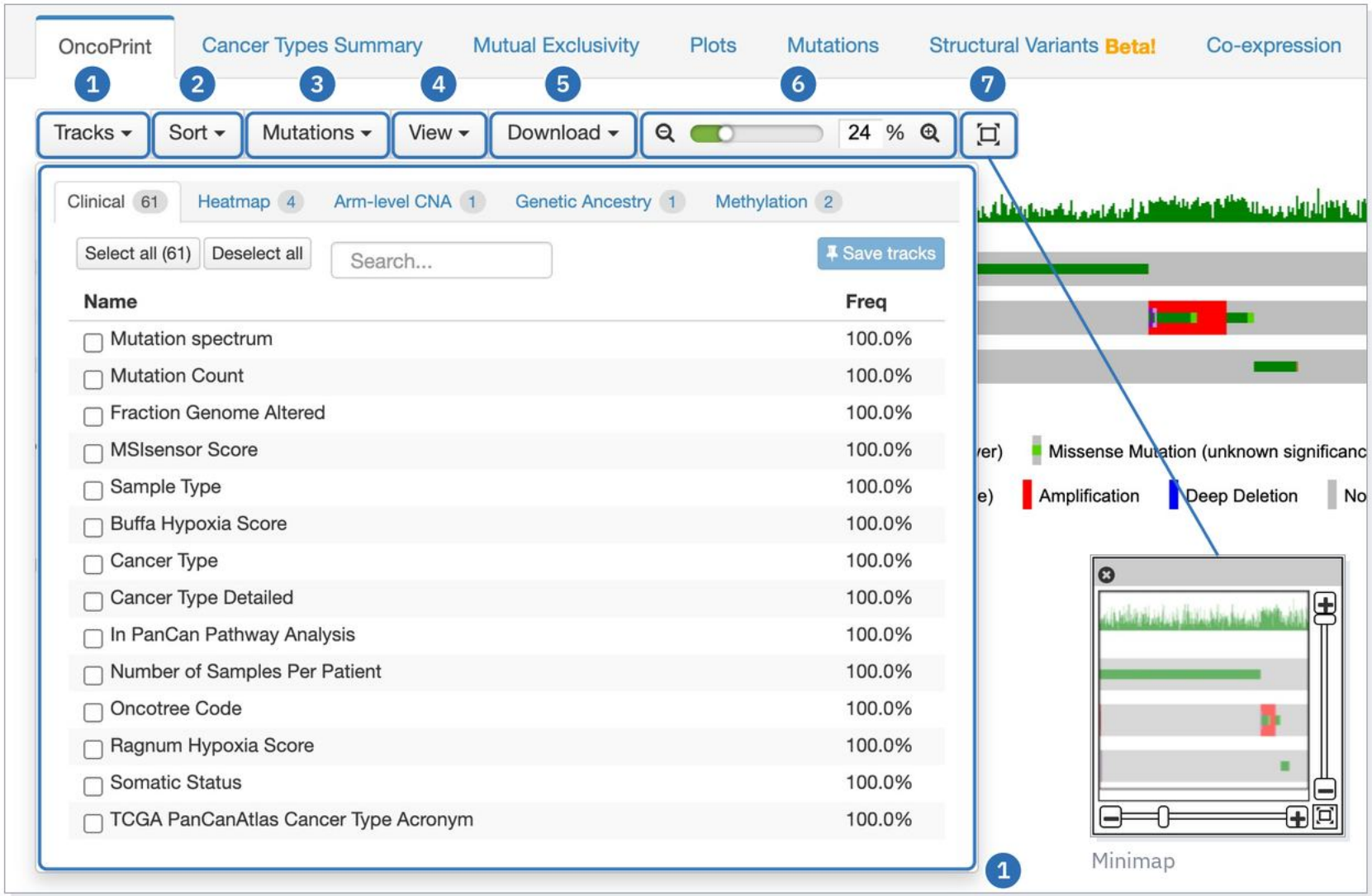
OncoPrint



1. Each column is a patient, each row a query gene. Colors show the kind of alteration; samples are sorted by them.
2. Share of queried samples altered in each gene: IDH1 77%, EGFR 11%, IDH2 5%. Drag a gene name to reorder the rows.
3. A clinical track, here Diagnosis Age. Add more from the “Tracks” menu.
4. Legend. Dark colors are putative drivers, light colors VUS (see the settings menu).

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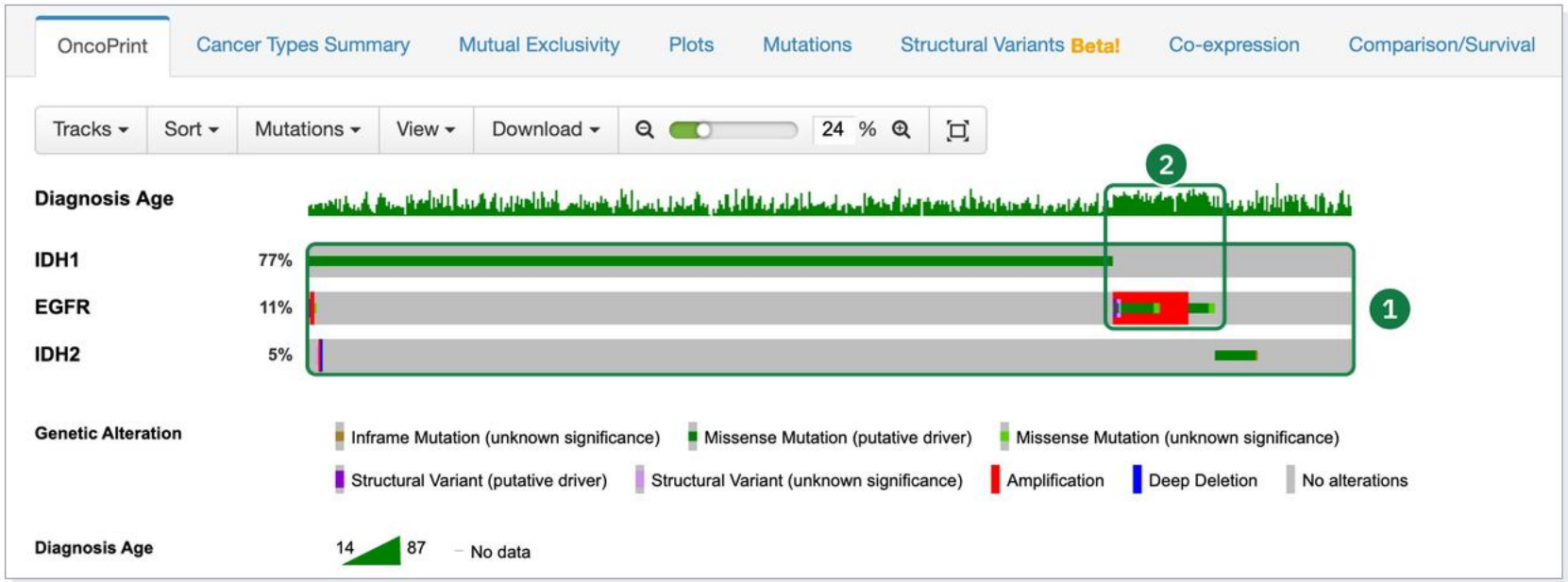
OncoPrint: the toolbar



1. "Tracks": add clinical attributes, heatmaps (e.g. mRNA), arm-level CNA, genetic ancestry or methylation. Tick attributes in the menu; "Save tracks" keeps them (only when logged in).
2. "Sort": by data (mutation type, driver), by case ID, or by case list order.
3. "Mutations": color by type, or somatic vs germline (only a few studies in the public portal have germline data).
4. "View": options for how the OncoPrint is laid out and what it shows.
5. "Download": save the OncoPrint as an image or its data, or open it in other tools.
6. Zoom out or in, or type a percentage.
7. The minimap: the whole OncoPrint in small. Drag or resize its window to move or zoom.

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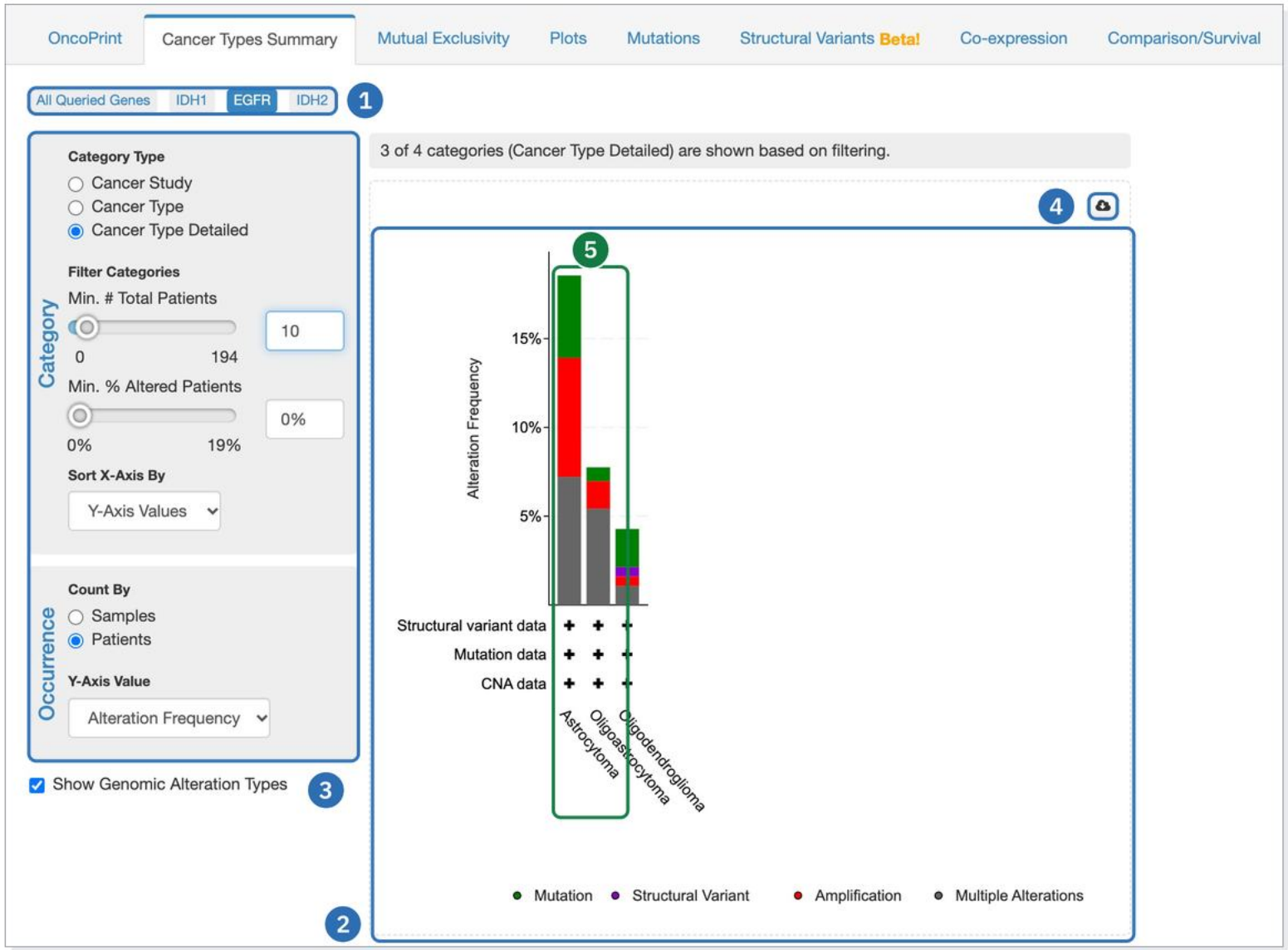
OncoPrint: Example uses



1. Q: Are alterations in these genes mutually exclusive?
   A: They seem to be: a sample altered in one gene is rarely altered in another. The Mutual Exclusivity tab tests it.

2. Q: Is age associated with alterations in a query gene?
   A: Patients with EGFR alterations look older (taller bars) than patients with IDH1 or IDH2 mutations. The Comparison/Survival tab tests it.

Cancer Types Summary



1. All query genes together, or one gene at a time (here EGFR).
2. Alteration frequency in each cancer type (here the detailed type), by kind of alteration. Hover over a bar for counts.
3. Options to group, filter and count the samples in the chart.
4. Download the chart.
5. Q: Are EGFR alterations more frequent in one subtype of glioma?
   A: Yes: in astrocytoma.

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Mutual Exclusivity



1. Test every pair of query genes for mutual exclusivity, co-occurrence or both; or list only significant pairs.
2. Samples altered in neither gene, in only one, or in both.
3. Log2 odds ratio: below 0, the genes tend to be altered in different samples; above 0, in the same samples.
4. p-value, and q-value adjusted for multiple testing. Hover over a column name for how it is calculated.
5. Q: Are alterations in IDH1, EGFR and IDH2 mutually exclusive, as the OncoPrint suggested?
   A: Yes for IDH1: it is mutually exclusive with EGFR and with IDH2 ($q < 0.001$). EGFR and IDH2 lean the same way, but not significantly ($q = 0.096$).

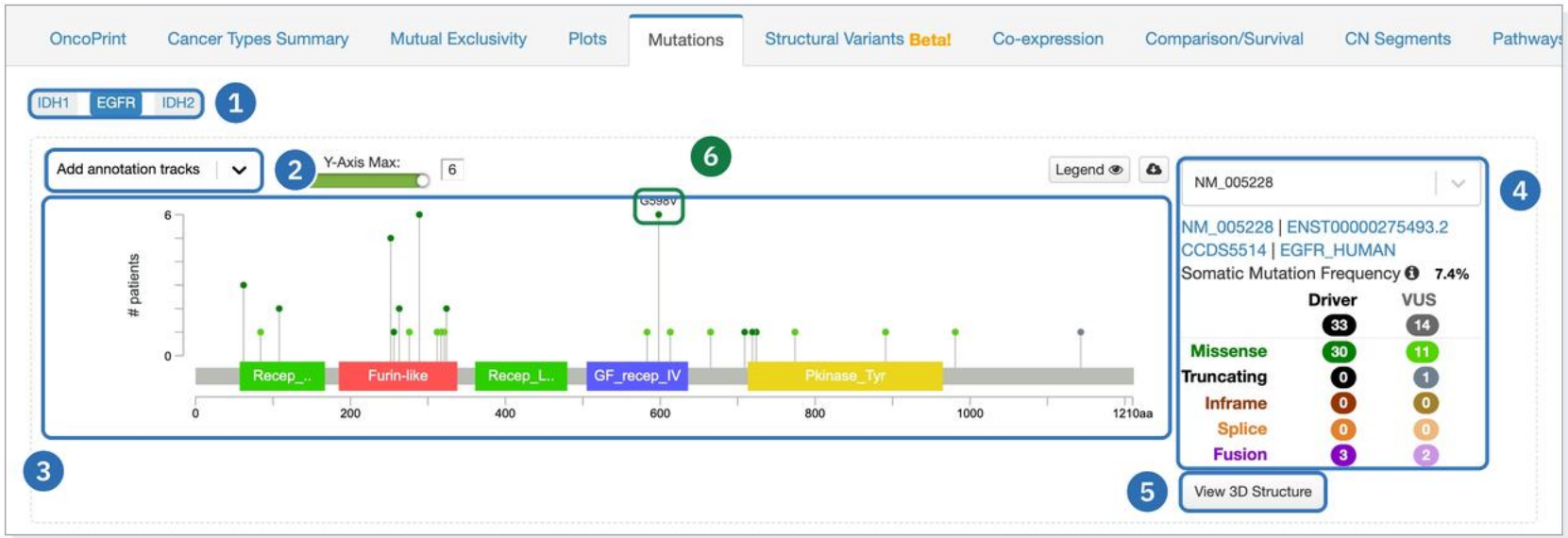
Plots



1. Choose what to plot on each axis below, or start from one of the example configurations (here mRNA vs CNA).
2. Horizontal axis: the data type (e.g. copy number, mRNA, clinical attribute), the profile and the gene.
3. Swap the axes.
4. Vertical axis: the same choices.
5. Color the samples by a gene's mutations, structural variants and copy number (here IDH1).
6. Each dot is a sample.
7. Download the plot.
8. Q: Does EGFR amplification change its expression?
   A: Yes: amplified samples have higher EGFR mRNA.

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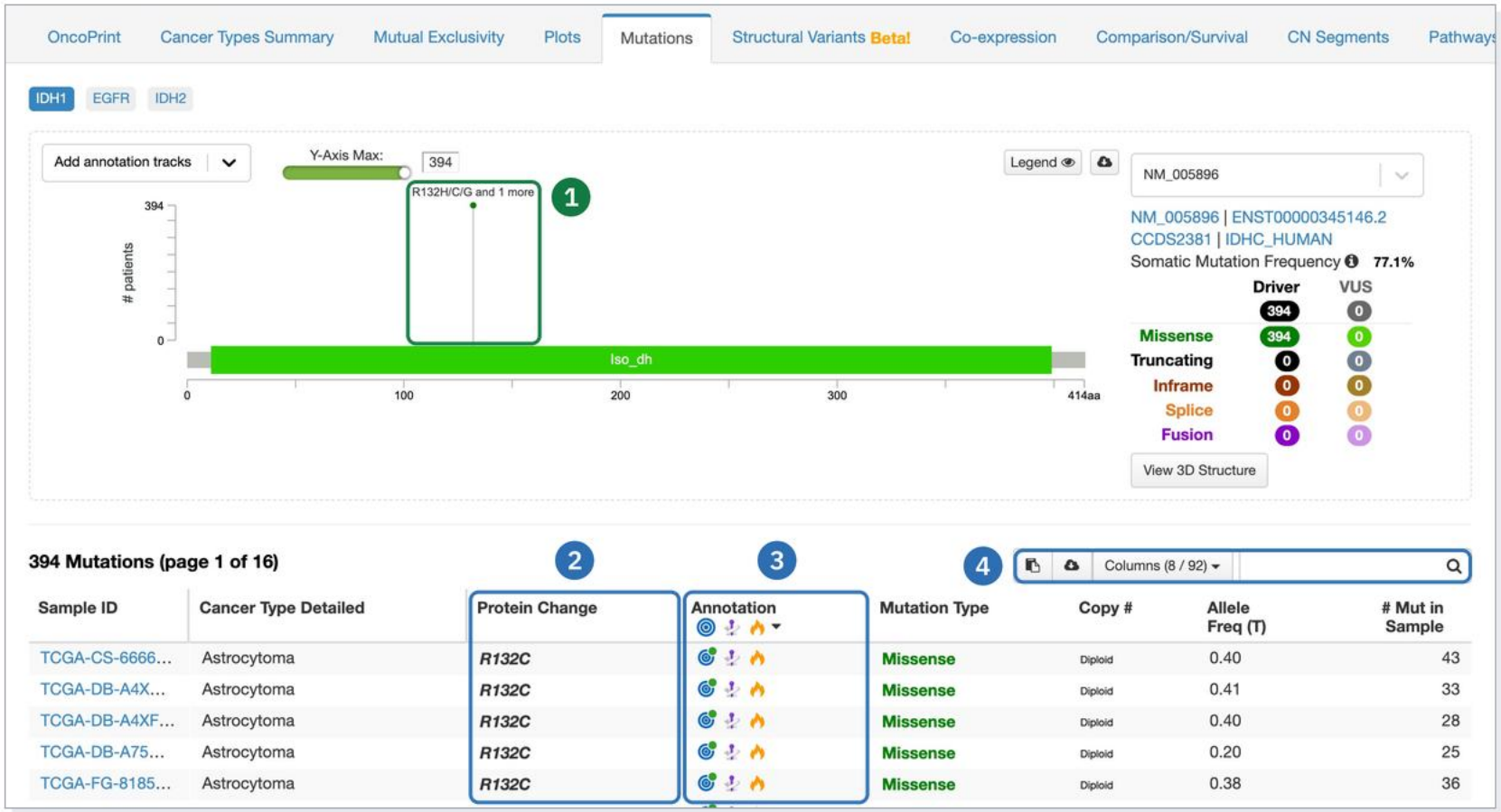
Mutations



1. One tab per query gene (here EGFR).
2. Annotation tracks: add tracks below the lollipop plot, such as Cancer Hotspots, OncoKB and post-translational modifications.
3. Each lollipop is a mutated position along the protein and its domains; its height is the number of samples. Hover for details, click to filter the table.
4. The transcript, and the number of driver and VUS mutations of each type.
5. View the mutations on the 3D protein structure.
6. Q: Where are the EGFR hotspots in glioma?
   A: G598V is one of the most frequent; most mutations fall in the extracellular domains (Recep_L, Furin-like, GF_recep_IV).

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Mutations: the table



1. Q: IDH1 mutations are highly recurrent. Are they known hotspots or drivers?
   A: Almost all are at R132 (R132H, R132C, R132G...).

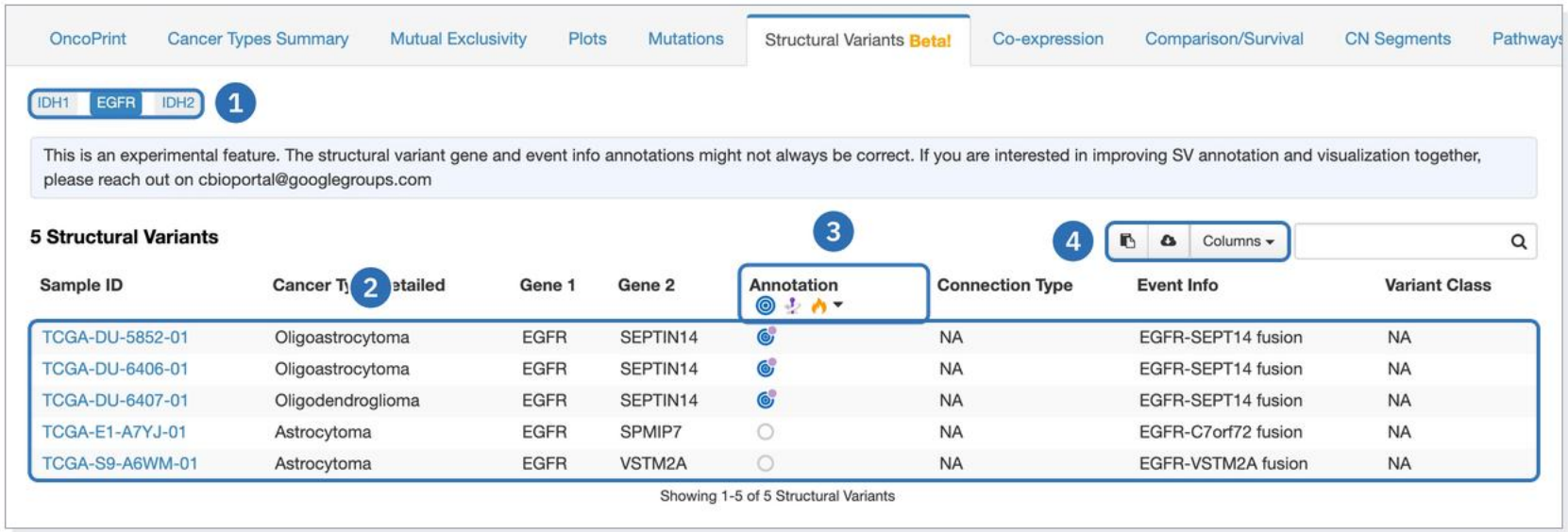
2. Protein change. Click a column header to sort; hover over it to filter that column.

3. Annotation: OncoKB (oncogenic, therapeutic levels), CIViC and Cancer Hotspots. Hover over an icon for details: R132 is a recurrent hotspot and a known oncogenic mutation.

4. Copy or download the table, add columns (8 of 92 shown), or search it.

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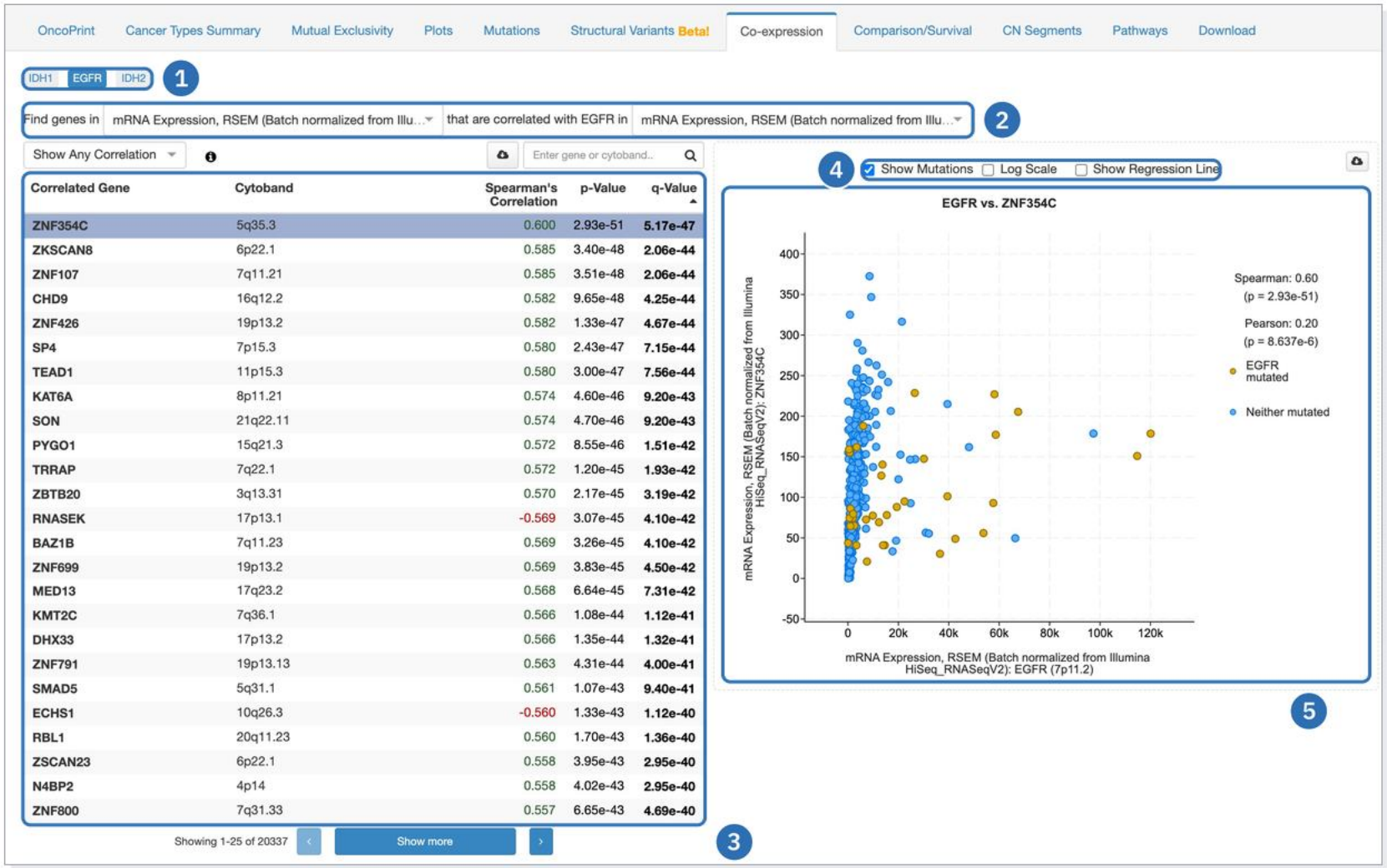
Structural Variants



1. One tab per query gene (here EGFR). A new tab (Beta): its annotations may not always be right.
2. One row per structural variant: the sample, the two genes and the event. Here, EGFR-SEPTIN14 fusions and others.
3. Annotation, as on the Mutations tab.
4. Copy or download the table, add columns.

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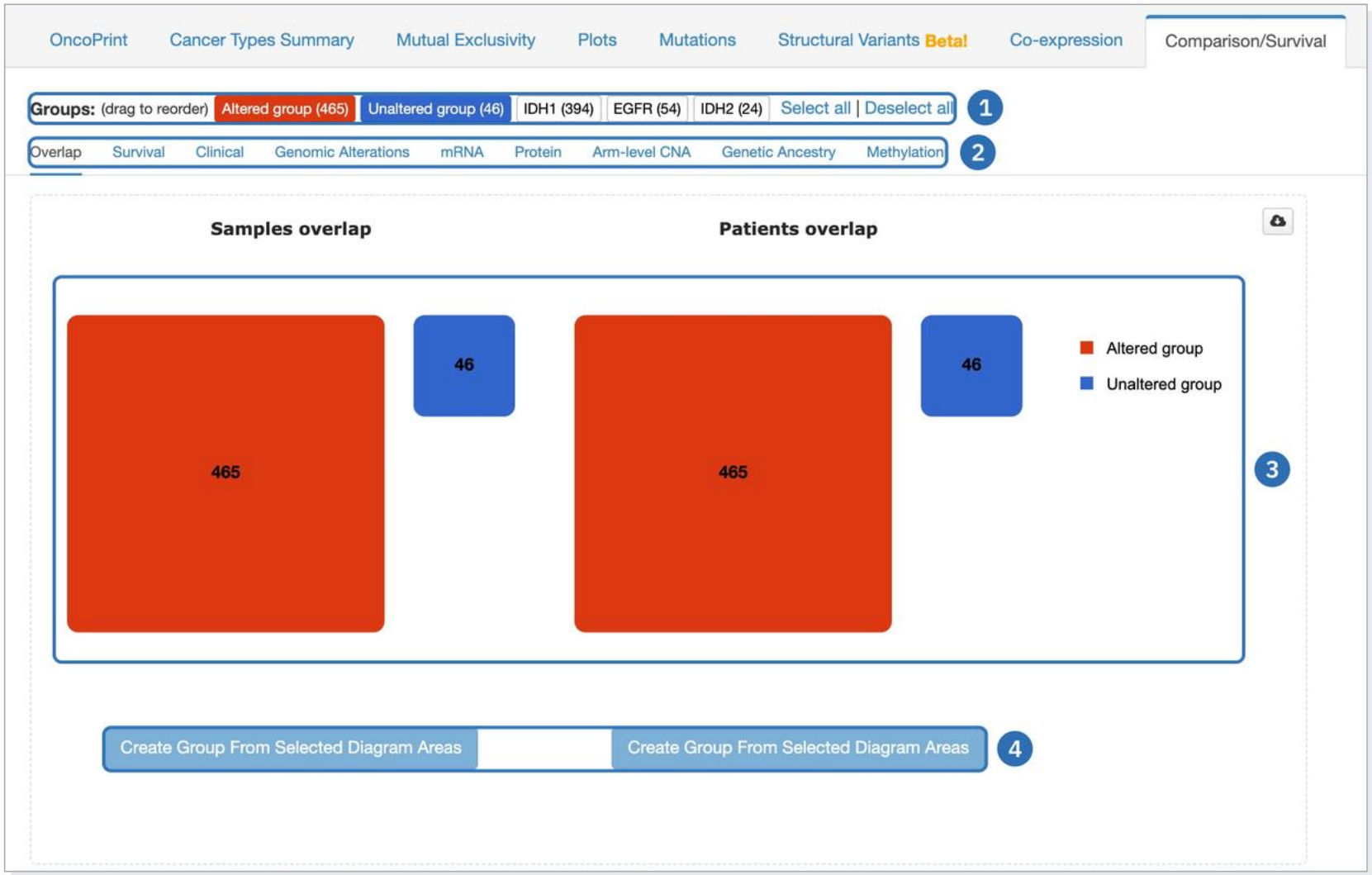
Co-expression



1. One tab per query gene (here EGFR).
2. Select profiles to compare expression of your query gene against all other genes.
3. Genes ranked by correlation with EGFR (Spearman, with p- and q-values). Click a gene to plot it.
4. Color by mutation status, use a log scale, add a regression line.
5. Each dot is a sample. Several of the top genes (ZNF107, TRRAP, BAZ1B, KMT2C) are on chromosome 7, like EGFR, which is often gained in glioma: see CN Segments.

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Comparison/Survival



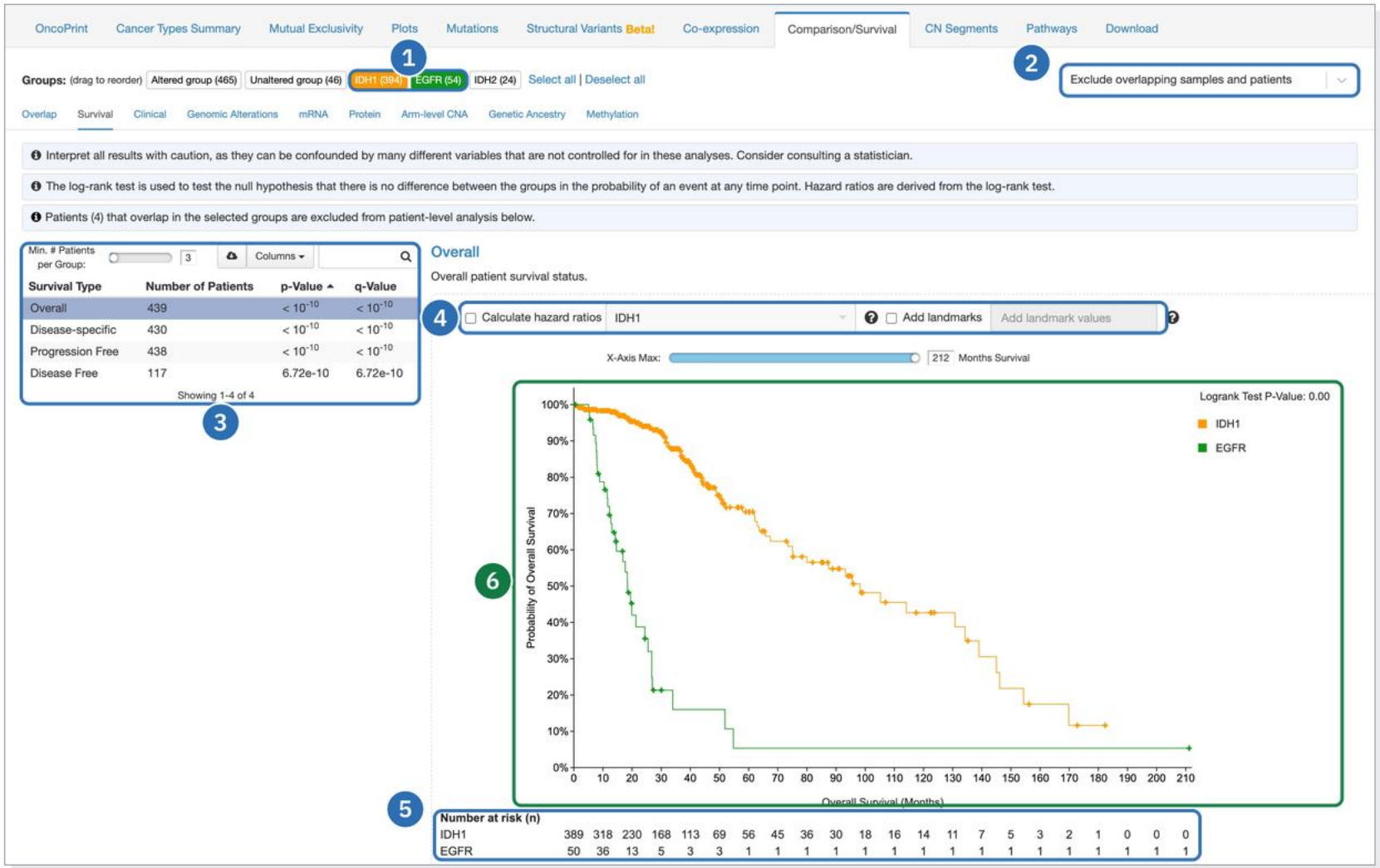
1. Groups to compare: Altered (465 samples with an alteration in any query gene) and Unaltered (46), plus one group per query gene. Click a group to select or deselect it.

2. Analyses, depending on the data: Overlap, Survival, Clinical, Genomic Alterations, mRNA, Protein and more. They work as on the Group Comparison page: see Tutorial 6.

3. Overlap: how the selected groups share samples and patients.

4. Select areas of the diagram to create a new group.

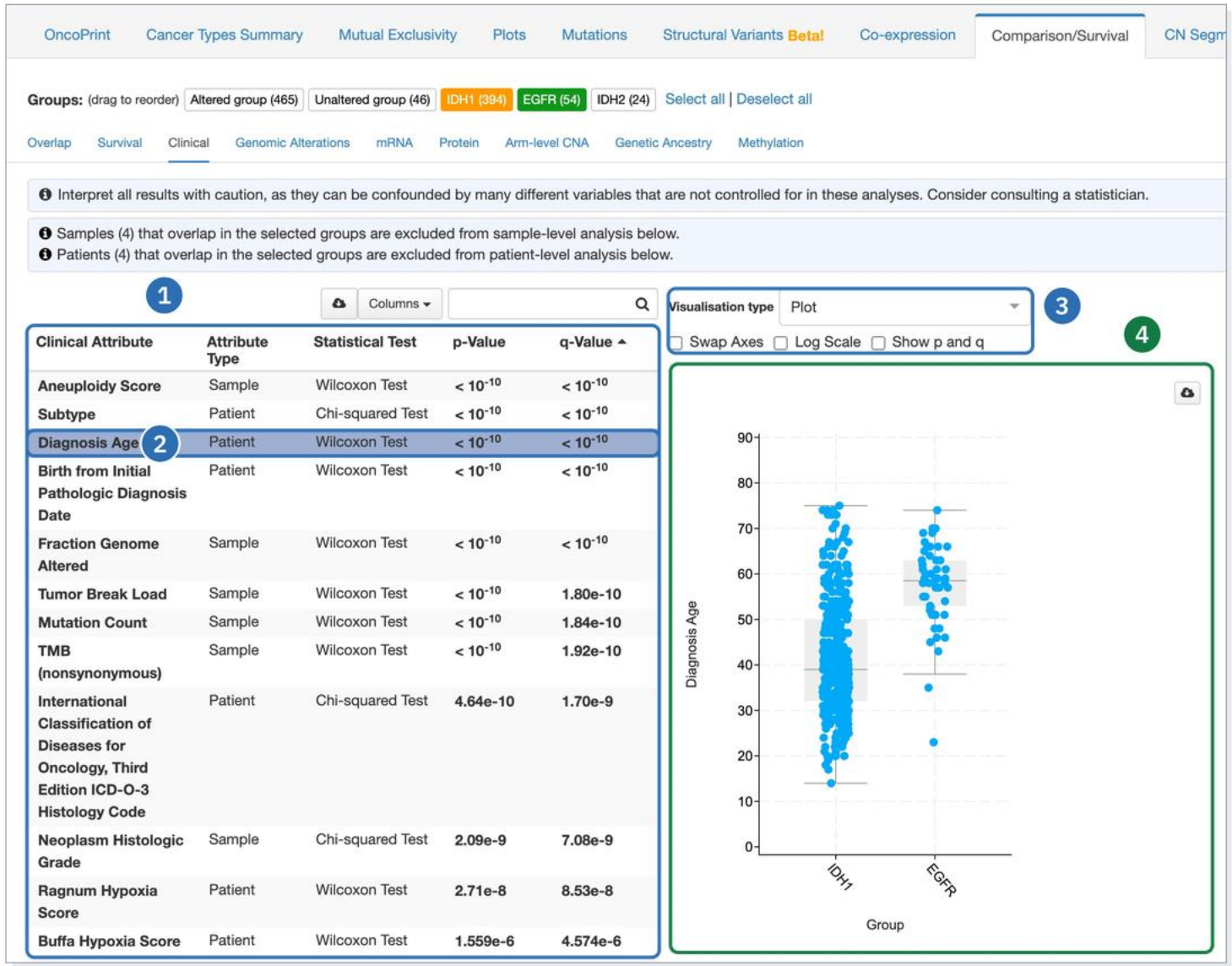
Comparison/Survival: survival



1. Here only the IDH1 and EGFR groups are selected.
2. Samples and patients in both groups are left out by default ("Exclude overlapping samples and patients"). Choose here to include them.
3. The outcomes in the study (overall, disease-specific, progression-free, disease-free), with log-rank p- and q-values. Click one to plot it.
4. Optionally calculate hazard ratios or add landmarks.
5. The number of patients at risk over time.
6. Q: Do patients with IDH1 or EGFR alterations have different outcomes?
   A: Yes: overall survival (the Kaplan-Meier curves) is much shorter with EGFR alterations.

Link to this page

Comparison/Survival: clinical



1. Every clinical attribute, tested between the groups, most significant first.
2. Click an attribute to plot it.
3. Options to customize the plot.
4. Q: Are patients with EGFR alterations older?
   A: Yes: diagnosis age is higher in the EGFR group (q < 1e-10).

Link to this page

Comparison/Survival: genomic alterations



1. The alteration types to test. All are ticked when the tab opens; untick some and click “Select” to test only the ones still ticked.

2. The results as a volcano plot, as frequencies in each group, and as the most frequently altered genes.

3. Every gene, tested for enrichment in each group (sample or patient level).

4. Tick genes and add them to the query (“Add checked genes to query”).

5. Q: Do samples without IDH1, IDH2 or EGFR alterations carry other alterations?
   A: NF1 is altered in 33% of them vs 4% of the altered group ($q < 0.001$). Add NF1 to the query and look at OncoPrint and Mutual Exclusivity.

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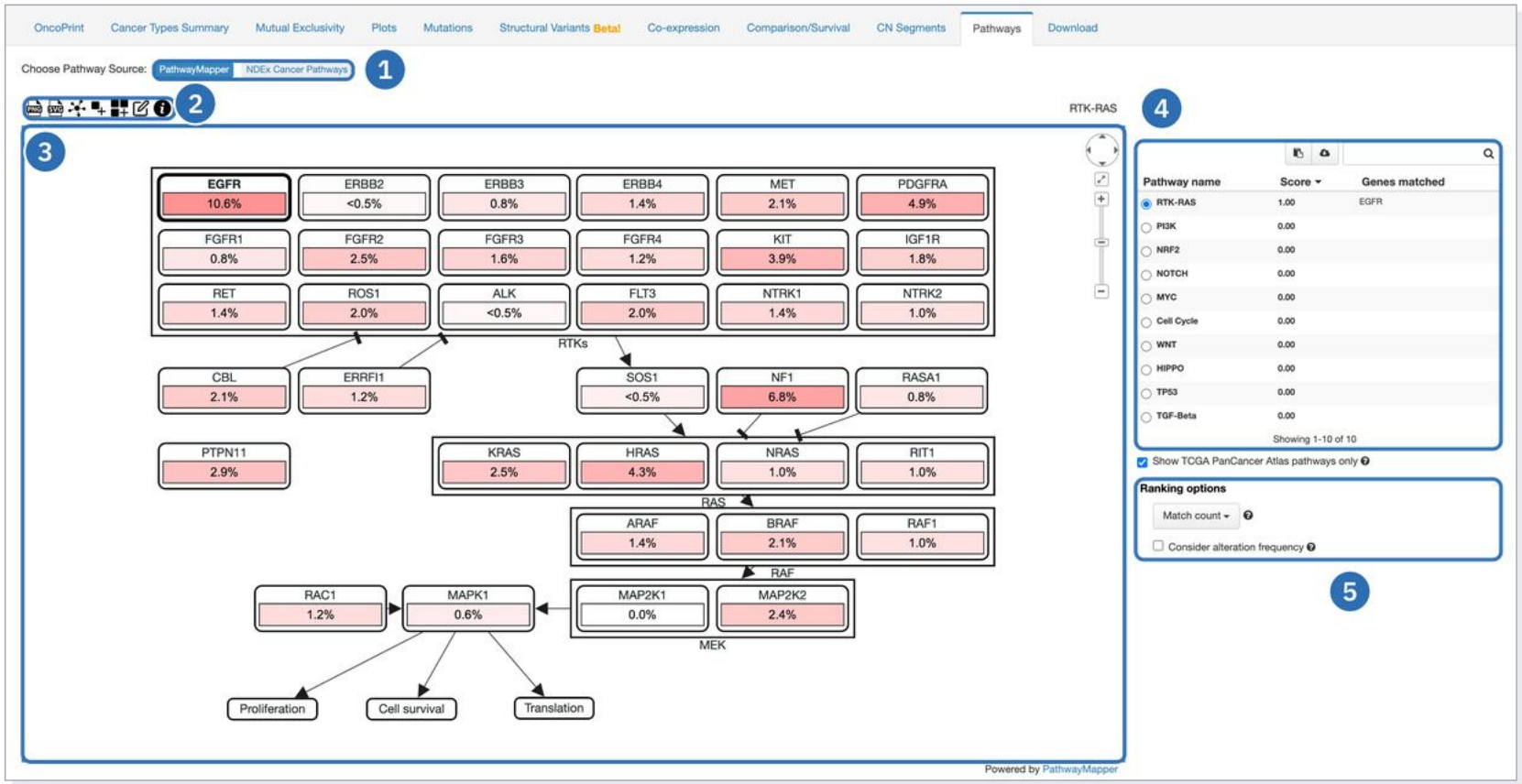
CN Segments



1. The whole genome, or one tab per query gene (here EGFR).
2. The Integrative Genomics Viewer (IGV): type a gene or coordinates to move there.
3. Options for the IGV display, and saving it as SVG.
4. Zoom out or in.
5. One row per sample: red for gains and amplifications, blue for losses.
6. Q: Are EGFR amplifications focal or broad?
   A: Zoom out to see.

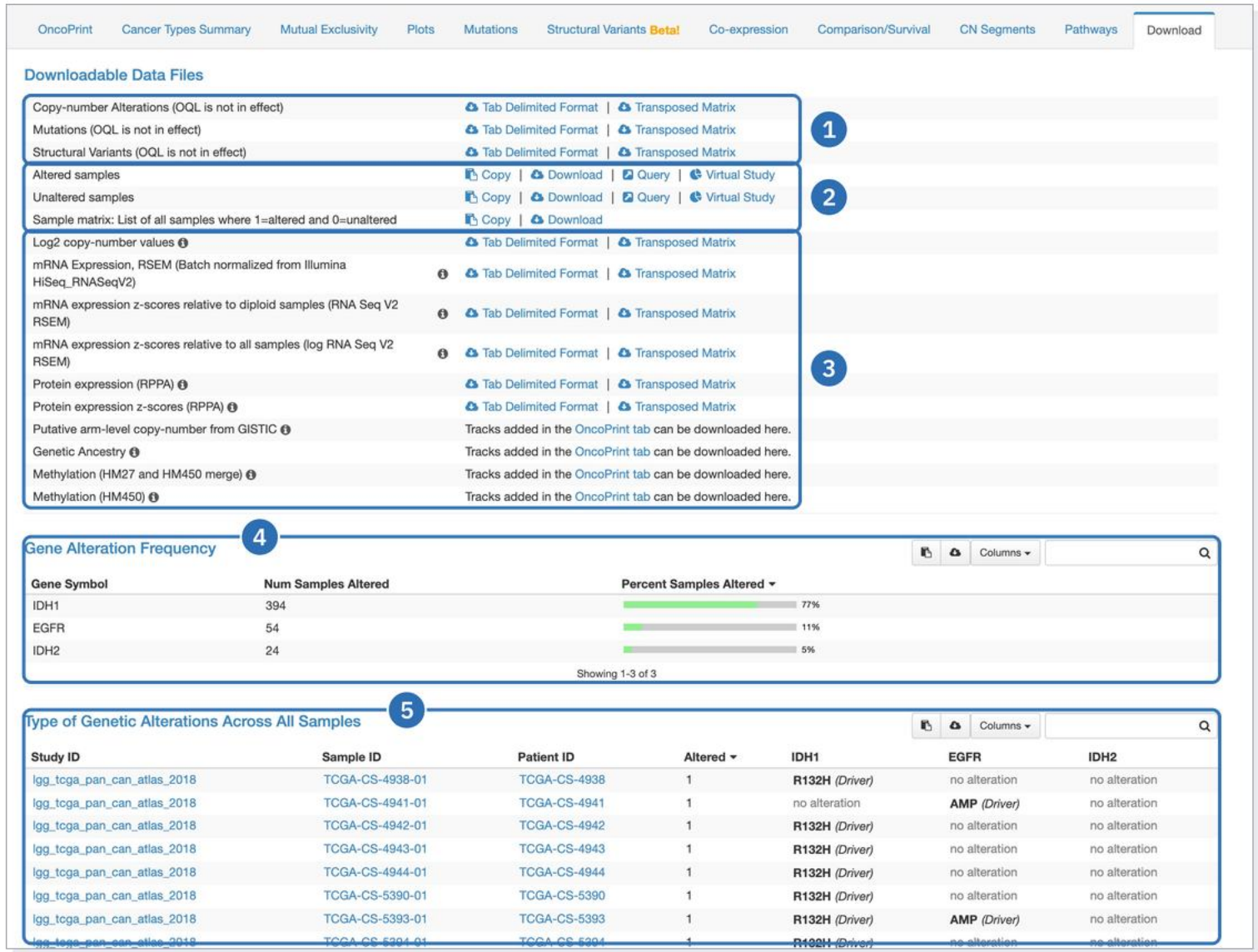
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Pathways



1. Pathways from PathwayMapper or NDEx.
2. Download (PNG, SVG) and edit the pathway; the info button explains the tools.
3. Each gene shows how often it is altered in the queried samples; query genes have a bold border (EGFR).
4. Pathways ranked by how many query genes they contain. Pick one to show it. See Tutorial 7.
5. Ranking options.

Download



1. The queried data (copy number, mutations, structural variants) for the query genes: a table or a transposed matrix.
2. Altered and unaltered samples: copy or download the list, query them, or save them as a virtual study.
3. The study's other profiles for the query genes. Tracks added in the OncoPrint are downloaded there.
4. Alteration frequency per gene.
5. Every queried sample and its alterations in the query genes.



Questions?

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